Yilin Wang

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Research Interests

- **LLM/VLM agents & world models for AV & physical AI** – mobility planning foundation models for sim agents, world-model construction, and safety-critical scenario generation & closed-loop validation.
- **Robotaxi & AV fleet operations** – routing, dispatching, and repositioning at network scale, with fleets doubling as mobile sensing platforms; via MILP and learning-based combinatorial optimization.
- **Agentic system implementation & deployment** – multi-agent architectures, tool orchestration, and agent evaluation for production workflows, from smart-city & robotaxi operations to automated research pipelines.
- **Cooperative perception & V2X systems** – from algorithms to real-world field deployment.

Education

- **Purdue University** *Aug.2021 - May 2027 (Expected)*
  Doctor of Philosophy West Lafayette, US
  - Advisor: Dr. Yiheng Feng. Dissertation (working title): *From Traffic Monitoring to Driving Behavior Reconstruction*.
  - Research Area: Connected and Automated Vehicle, Cooperative Perception, Physics-informed Machine Learning, Robotaxi Operation and Control
  - GPA: 3.8/4.00

- **Carnegie Mellon University** *Aug.2019 - Dec.2020*
  Master of Science, Advanced Infrastructure Systems Pittsburgh, US
  - Area: Smart City, Deep Learning, Natural Language Processing
  - Course: Deep Learning, Reinforcement Learning, Geo-Informatics System
  - GPA: 3.87/4.00

- **Tongji University** *Aug.2015 - Jun.2019*
  Bachelor of Science Shanghai, China
  - Major: Civil Engineering (Area of Geo-technical and Underground Engineering)

Patents and Publications

C=Conference, J=Journal, S=In Submission, P=In Preparation

[S.1] **Yilin Wang**, Yiheng Feng* (2026). RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring. Under review by *Transportation Research: Part C*. Preprint: arXiv:2609.06813.

[P.1] Neural Force: Interpretable Learned Car-Following Response Fields. Manuscript in preparation (co-author).

[J.1] Hanlin Chen, Vamsi K Bandaru, **Yilin Wang**, Mario A Romero, Andrew Tarko, Yiheng Feng* (2024). Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety in *Transportation Research Record*, 2678(12), 1498-1510. https://doi.org/10.1177/03611981241252779

[J.2] **Y. Wang** and Y. Feng* (2024). IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction in *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 6, pp. 5014-5020, June 2024, doi: 10.1109/TIV.2024.3367654

[J.3] Ya-Dong Xue, Wei Zhang, **Yi-Lin Wang***, et al. (2023). Serviceability evaluation of highway tunnels based on data mining and machine learning: A case study of continental United States in *Tunnelling and Underground Space Technology*, Volume 142, 2023, 105418, ISSN 0886-7798, https://doi.org/10.1016/j.tust.2023.105418

[C.1] **Yilin Wang**, Yiheng Feng* (2026). RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring. In *INFORMS 2026 Annual Meeting*, Accepted with Oral Presentation.

[C.2] **Yilin Wang**, Yiheng Feng* (2026). A Cooperative Perception Based Dynamic Vehicle Routing Framework for Urban Traffic Monitoring. In *Transportation Research Board Annual Meeting 2026*, Accepted by Poster Presentation.

[C.3] **Yilin Wang**, Yiheng Feng* (2024). Dynamic Routing of Connected and Automated Vehicles for Improving Network Coverage. In *Transportation Research Board Annual Meeting 2024*, Accepted by Poster Presentation.

RESEARCH PROJECTS

- **TeraSim: City-Scale World Model for Safety Testing on AV and Physical AI** *Dec.2025 - Present*
  Core Developer | Internship | SaferDrive AI(Inchor)
  - **Overview**: A city-scale simulation platform serving as the foundational infrastructure for agentic world models, enabling naturalistic and adversarial testing of autonomous vehicles at urban scale.
  - **Core Module – NNDE (Neural-network Naturalistic Driving Environment)**: Behavioral-layer foundation module of TeraSim that integrates a transformer-based motion model to drive city-scale, naturalistic AV behavior generation.
  - **Milestone**: 2026-03: Active deployment with NVIDIA and City of San Jose as an Agentic Incident-to-Operation pipeline, bringing TeraSim from simulation into real-world city operations.
  - **Tech Stack**: Python, SUMO, PyTorch, CUDA, Docker, FastAPI

- **AgentSim: An Agentic Test Engineer for Autonomous-Driving Safety Validation** *Mar.2026 - Present*
  Initiator | Internship | SaferDrive AI(Inchor) [Demo]
  - **Overview**: An end-to-end, LLM/VLM-driven multi-agent platform that generates safety-critical scenarios with trajectories or text input.
  - **Milestone**:
    1. 2026-06: First draft release of AgentSim, utilizing LLMs and VLMs for safety-critical scenario generation.
    2. 2026-10: Public AgentSim demo: 3 real drives, each with 5 generated safety-critical variants rendered as synchronized front-camera video.

- **Safety Instinct for Physical AI: A CBF-Based Safety Cerebellum for Quadruped Robots** *Feb.2026 - Apr.2026*
  Independent Research | Internship | SaferDrive AI(Inchor)
  - **Overview**: An internal exploration of a "safety instinct" for Physical AI – a safety cerebellum for a Boston Dynamics Spot robot, built in NVIDIA Isaac Sim (Omniverse).
  - **Methodology**: Trained a reinforcement learning (RL) waypoint-following controller for Spot, then layered a Control Barrier Function (CBF) safety filter beneath it – the learned policy handles navigation while the CBF layer enforces safety.
  - **Contribution**: Enabled Spot to maintain a dynamic safety boundary around moving obstacles while traveling between waypoints; project conceived and completed independently.
  - **Tech Stack**: NVIDIA Isaac Sim (Omniverse), Reinforcement Learning, Control Barrier Functions

- **Cooperative Perception CAV Fleet Operation for Data Collection** *Apr.2023 - Jun.2026*
  Individual Research Project [GitHub]
  - **Project 1**: Dynamic Routing on Connected and Automated Vehicles for Improving Network Coverage Apr.2023 - Aug.2024
    - **Research Question**: How to model the CAV network coverage and operate routes to improve data collection.
    - **Contribution**:
      1. We formulate the CAV routing problem considering the network coverage as one objective.
      2. We propose a heuristic algorithms with greedy search to solve the multi-objective optimization.
      3. We conduct a simulation experiment in a grid network to verify the feasibility and relations between travel cost and network coverage.
    - **Methodology**: heuristic optimization, SUMO simulation, link-based travel time estimation.
    - **Outcome**: Accepted by Transportation Research Board (TRBAM) 2024 as poster presentation.
  - **Project 2**: RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring Sept.2024 - Jun.2026
    - **Research Question**:

1. A higher granularity of road network representation is needed.
2. A proper performance metric is needed to evaluate traffic monitoring impact.
3. There is a dynamic effect between CAV routes decision and partial network observations.

* **Contribution**:
1. We apply Cell Transmission Model (CTM) for traffic state prediction and road network representation considering CAV detection maneuver.
2. We propose a comprehensive MILP formulation considering the performance of traffic monitoring as well as dynamic vehicle routing in one explicit equation.
3. We conduct a microscopic simulation and evaluations under various scenarios.

* **Outcome**: Under review by Transportation Research: Part C; preprint arXiv:2609.06813; poster at TRB 2026; accepted for oral presentation at INFORMS 2026.

- **Modeling Car-Following Behavior with Physics-Informed Neural Network** *Jan.2022 - Present*
Behavior-level reconstruction of human driving: from physics-constrained prediction to cognitively interpretable response [GitHub]
  - **Overview**: Reconstruct human driving behavior by embedding physical and cognitive structure into neural networks, so that learned behavior models stay accurate, robust, and interpretable – the behavior layer behind sim agents and world models.
  - **Project 1**: IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction Jan.2022 - Feb.2024
    * **Research Question**: How to integrate an explicit, physics-based car-following model into a learning-based predictor to improve training and performance.
    * **Contribution**:
      1. We introduce a physics-informed neural network (PINN) model that integrates the Intelligent Driver Model (IDM) into a neural trajectory predictor.
      2. The proposed model exhibits robustness against real-time GPS noise.
    * **Methodology**: A customized attention-based VAE trained with a loss that combines physical-law residuals and ground-truth error.
    * **Outcome**: Published in IEEE Transactions on Intelligent Vehicles (first author); research presentation at IEEE-IV 2024, Jeju Island, KR.
  - **Project 2**: Neural Force: Interpretable Learned Car-Following Response Fields
    * **Research Question**: How does a driver combine perceptual cues – spacing, closing speed, and own speed – into one acceleration decision, and can a learned model keep that combination readable instead of hiding it in a black box?
    * **Contribution**:
      1. We propose a cognitively motivated response-field model: acceleration is assembled from signed, weighted gradients of one shared learned scalar, and each term carries a behavioral role (restore spacing, oppose closing, damp speed).
      2. We derive interpretable readouts – operating gap, gap-restoration gain, and closing-opposition gain – that make learned human behavior comparable across datasets.
      3. Across four naturalistic corpora (AD4CHE, highD, LA Highways, I-24 MOTION), we find a shared corrective structure but distinct sensitivities: five of six corpus pairs reverse order between the two gains, ruling out a single "response strength" axis.
    * **Methodology**: Neural scalar potential with state-dependent convex weights; Student-t residual for uncertainty; annealed IDM prior during training.
    * **Outcome**: Manuscript in preparation (co-author).

- **Developing A Cooperative Perception System** *Jan.2023 - Present*
Project Leader | Funded by Center of Connected and Automated Transportation
  - **Phase 1**: A Cooperative Perception System for CAV Navigation and Improving Safety
    * **Objective**: Implement a cooperative perception system to improve road safety.
    * **Contribution**:

1. We build a cooperative perception system connected by V2X between roadside LiDAR and CAV sensors.
2. We develop a detection and navigation algorithm on the CAV with a sensor data fusion pipeline.
3. We build and conduct field testing demonstrating the safety benefits of detecting occluded VRUs through the cooperative perception system.

* **Conclusion**: The cooperative perception system helps CAVs plan safer (i.e., higher post-encroachment time) and smoother (i.e., lower deceleration rates) trajectories.
* **Outcome**: Publication [J.1] in Transportation Research Record.

◦ **Phase 2**: An Infrastructure-Vehicle Cooperative Perception System

* **Objective**: Develop a cooperative perception system including multiple vehicles, infrastructure sensors, and drones.
* **Contribution**: Extract and process multi-sensor data (LiDAR, camera, etc.) into vehicle trajectories; release an open-source dataset for future research.
* **Methodology**: Multi-sensor fusion of infrastructure sensors, drone-view cameras, and vehicle sensors.

• **Developing V2X System for Traffic Signal Control** *Sept.2021 - Jun.2026*
Multiple Projects, Funded by US-DOT, Michigan DOT, and Leidos Inc.

◦ **Part 1: Smart Intersection Project** Sept.2021 - Jun.2026

* **Overview**: Implement CV2X system for Transit Signal Priority (TSP) in City of Ann Arbor.
* **Contribution**:
  1. Redevelop Multi-Modal Intelligent Traffic Signal System (MMITSS). [GitHub]
  2. Maintain and update versions on road-side and vehicle-side applications.
  3. Field testing, installation, and data processing for buses and real intersections. [GitHub]

◦ **Part 2: Signal-Aware Speed Advisory on a Connected and Automated Vehicle**

* **Overview**: A full-stack, closed-loop infrastructure-to-AV system that plans the vehicle trajectory in real-world deployment.
* **Contribution**:
  1. Roadside: connected the traffic-signal controller to roadside V2X unit.
  2. Vehicle side: Receive and decode road-side information, execute the trajectory with a signal-aware speed advisory planner.
  3. Deployed and validated the full pipeline on a real vehicle, closing the loop from signal to vehicle motion.
* **Outcome**: Adopted as the class demo of Purdue CE 565 Traffic Engineering.

◦ **Part 3: CDA-Sim Update: Integrating MMITSS on a Co-simulation Platform** Dec.2024 - Jun.2025

* **Overview**: Integrating MMITSS into CARMA-CARLA-SUMO co-simulation.
* **Contribution**:
  1. New features on time synchronization, message brokers, and parallel processing to support MMITSS in CDA-Sim platform. [GitHub]
  2. Build integration testing and demo with released configuration.

◦ **Part 4: Michigan-DOT ATCMTD Project**

* **Overview**: Implementing the MMITSS system for the Q-Line in Detroit, MI.
* **Contribution**: Updated MMITSS to accommodate the NTCIP 1211 operation protocol. [GitHub]

WORKING EXPERIENCE

• **Inchor(SaferDrive AI)** *Dec.2025 - Jun.2026*
Simulation Development Engineer Intern Ann Arbor, MI

◦ **TeraSim – Naturalistic Driving Behavior**

* Developed and maintained TeraSim, an open-source city-scale traffic simulation platform for AV safety testing.
* Led integration of MotionTransformer as the foundation model of NNDE, generating naturalistic driving behavior at city scale alongside the NADE adversarial-scenario module; refactored the behavior-model pipeline from a dual-state design into a stateless prediction pipeline for large-scale inference.

    - Implemented a full-stack Driving Foundation Model into product; led the joint project with NVIDIA and the City of San Jose on agent-driven Smart City operation.
  - **AgentSim – Agentic Synthetic Scenario Generation** [Demo]
    - Provide an end-to-end pipeline: from trajectory/text description to safety-critical scenarios with ego camera-view data.
    - Develop an agentic platform with LLMs/VLM agents to collaborate, reconstruct, mutate, and render driving scenarios.

- **Purdue University** *Fall 2021 - Present*
  Research Assistant, supervised by Prof. Yiheng Feng West Lafayette, IN
  - Research Area: Connected and Automated Vehicle, Cooperative Perception, V2X Communication.

- **Carnegie Mellon University** *Jan.2020 - Dec.2020*
  Research Assistant, supervised by Prof. Pingbo Tang Pittsburgh, PA
  - Research Area: Natural Language Processing, Smart City.

Teaching Experience

- **Teaching Assistant – CE 361: Transportation Engineering** *Spring 2023, Spring 2024*
  Purdue University, Lyles School of Civil & Construction Engineering West Lafayette, IN
  - Assisted in teaching undergraduate transportation engineering courses covering traffic flow theory, highway design, and transportation planning.

- **Guest Lecturer – CE 565: Traffic Engineering** *Fall 2021 - Fall 2026*
  Purdue University, Lyles School of Civil & Construction Engineering West Lafayette, IN
  - Delivered guest lectures on SUMO (Simulation of Urban Mobility) traffic simulation for graduate-level traffic engineering course.

Peer Review & Service

- **Transportation Research Part C: Emerging Technologies** *Journal Reviewer* *2024, 2025*
- **IEEE Transactions on Intelligent Vehicles** *Journal Reviewer* *2024, 2025*
- **Journal of Intelligent Transportation Systems** *Journal Reviewer* *2023-2025*
- **Multimedia Tools and Applications** *Journal Reviewer* *2023-2025*
- **IEEE Intelligent Transportation Systems Conference (ITSC)** *Conference Reviewer* *2024*
- **IEEE-IV24 Workshop on Socially Interactive Autonomous Mobility** *Committee Member* *2024*

Skills

- **Programming Languages:** Python, C++, MATLAB, JavaScript
- **Data Science & Machine Learning:** PyTorch, CUDA, Hugging Face
- **Simulation & Optimization:** SUMO, CARLA, CARMA, NVIDIA Isaac Sim, Gurobi
- **Cloud Technologies:** AWS, SSH
- **DevOps & Version Control:** Linux, Docker, Conda
- **Cooperation Platform & Project Management:** Jira, Slack, GitHub
- **Other Tools & Technologies:** Microsoft Office, Auto-CAD, ArcGIS
- **Research Skills:** Critical Thinking, Research Talk, Comprehensive Writing, Literature Review

Honors and Awards

- **Honorable Mention** Mathematical Contest in Modeling of USA *2018*

Additional Information

Languages: English (Proficiency), Chinese Mandarin (Native)
Interests: Soccer, Skateboarding, Range and Gaming

REFERENCES

1. **Yiheng Feng**
   Associate Professor, Civil and Construction Engineering
   Assistant Director, Center for Road Safety (CRS)
   Lyles School of Civil and Construction Engineering, Purdue University
   Relationship: Ph.D Advisor

2. **Pingbo Tang**
   Associate Professor, Civil and Environmental Engineering
   Carnegie Mellon University
   Relationship: Master Research Advisor

Contact details for references available upon request.